MATH 205

Linear Algebra

LECTURE 1 · SYSTEMS OF LINEAR EQUATIONS

Where every idea in this course begins: equations, elimination, and the shape of a solution set.

Second-Year CS & Engineering Core Math

Texts: Anton & Rorres, Elementary Linear Algebra, 9th Ed. · Margalit & Rabinoff, Interactive Linear Algebra

WELCOME

Before We Start

Texts

Primary: Anton & Rorres, Elementary Linear Algebra, 9th ed.

Reference: Margalit & Rabinoff, Interactive Linear Algebra (free, online)

How this course runs

- Lecture builds intuition first, formalism second
- Every definition gets a worked example the same day
- Homework from Anton & Rorres exercises weekly

Where Today Fits

1. **Systems of Linear Equations** — YOU ARE HERE
2. Matrices & Matrix Algebra
3. Determinants
4. Vector Spaces
5. Eigenvalues & Eigenvectors

MOTIVATION

Why This Course Matters to You

Every one of these is, underneath, a system of linear equations.

Computer Graphics

Rotating, scaling, and projecting 3D models — every transform is a matrix acting on coordinate vectors.

Machine Learning

Linear regression, PCA, and the core of neural network layers all reduce to solving or approximating Ax = b.

Circuit Analysis

Kirchhoff's laws turn a resistor network into a linear system whose unknowns are the branch currents.

Search & Ranking

Google's PageRank finds the dominant eigenvector of a giant linear system built from the web's link graph.

Robotics & Kinematics

Relating joint angles to end-effector position is a linear (or linearized) system, solved every control cycle.

Structural Engineering

Truss and frame analysis solves a linear system for internal forces under load.

DEFINITION

What Is Linear Algebra?

Linear Algebra: the branch of mathematics concerned with linear equations, matrices, determinants, and vector spaces.

linear → relating to the **first degree**; having no variable raised to any power (other than 1), and no products of variables.

We will spend the whole semester on objects built from exactly this idea — starting today with equations themselves.

The Linear Equation

A line in the xy-plane is an equation of the form

$$a_1x + a_2y = b$$

More generally, a linear equation in the n variables x_1, x_2, …, x_n is one that can be written

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

a_1, a_2, …, a_n and **b** are real constants called **coefficients**. The variables are called the **unknowns**.

A solution is a set of numbers $\{s_1, s_2, \ldots, s_n\}$ that satisfies the equation when $x_i = s_i$ for every i.

QUICK CHECK

Is It Linear?

All variables to the first power. No products of variables. No variables inside functions.

$$x + 3y^2 = 7$$

No

y^2 has degree 2 — variables must appear only to the first power.

$$y - \sin x = 0$$

No

x sits inside a trig function — not first-degree.

$$3x + 2y - z + xz = 4$$

No

the term xz is a product of two variables.

$$\sqrt{x_1} + 2x_2 + x_3 = 1$$

No

$\sqrt{x_1}$ is a root of a variable, not a first-degree term.

Systems of Linear Equations

A system of linear equations *(or linear system)* is a finite set of linear equations in the same variables $x_1, x_2, \ldots, x_n$.

A general system of m equations in n unknowns is written:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

a_{ij}
first subscript i = equation number, second subscript j = which unknown it multiplies

m equations, n unknowns — m and n need not be equal.

A solution to the system is a set of values for $x_1, \ldots, x_n$ that satisfies every equation simultaneously.

PART TWO

What is a Matrix?

We've defined equations and systems. Now we give their coefficients their own object — and the rules for working with it.

The Matrix

Matrix: a rectangular array of numbers enclosed in brackets. The numbers in the array are called the **entries** of the matrix.

Example

$$A = \begin{bmatrix} 2 & 4 & 8 \\ 5 & -3 & 0 \end{bmatrix}$$

The plural of matrix is matrices.

We'll use capital letters (A, B, C, ...) for matrices and lowercase for their entries.

Size (Order) of a Matrix

The size of a matrix is described by the number of **rows** (horizontal lines) and **columns** (vertical lines): rows × columns.

$$A = \begin{bmatrix} 2 & 4 & 8 \\ 5 & -3 & 0 \end{bmatrix}$$

2 rows **3 columns**

size: 2 × 3

2 × 3 = 6 gives the total number of entries.

Convention: rows are always listed first, then columns — "rows by columns," always in that order.

This ordering matters — we'll lean on it constantly once we get to matrix multiplication.

Rectangular vs. Square Matrices

Rectangular Matrix

rows ≠ columns

$$A = \begin{bmatrix} 2 & 4 & 8 \\ 5 & -3 & 0 \end{bmatrix}$$

size 2 × 3 — 2 ≠ 3

Square Matrix

rows = columns = n — "order n"

$$B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

size 2 × 2 — order 2

Either way, we call the matrix an m × n matrix, or a matrix of order m × n.

Row and Column Vectors

Two special shapes come up constantly enough to earn their own name:

Column Vector

a matrix with only one column

$$\begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

size 2 × 1

Row Vector

a matrix with only one row

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix}$$

size 1 × 3

General Notation: The Entries a_{ij}

An m × n matrix is written with double-subscript entries:

$$B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix}$$

b_{ij}

the entry in row i and column j of B

first subscript = row

second subscript = column

(exactly the same convention we used for the coefficients aij in a linear system)

The Main Diagonal

In a square matrix A of order n, the entries $a_{11}, a_{22}, \ldots, a_{nn}$ lie on the **main diagonal**.

Diagonal matrix: a square matrix in which every entry off the main diagonal is zero.

Diagonal? (yes / no)

$$\begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$$

Yes

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Yes

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

No — not square

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

Yes — zero matrix

The Zero (Null) Matrix

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

A matrix whose entries are **all zero** is called the zero matrix (or null matrix), denoted **O**.

It plays the same role among matrices that 0 plays among numbers — we'll see that clearly once we define matrix addition.

Matrix Addition

If A and B are matrices of the **same size**, the sum A + B is obtained by adding corresponding entries.

Different sizes → the sum is undefined.

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 2 \\ 4 & -2 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 2 \\ 5 & 6 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

A + B is not defined — A is 3×3, B is 2×2.

B + C is defined — same size:

$$B + C = \begin{bmatrix} 3+1 & 2+0 \\ 5+0 & 6+1 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 5 & 7 \end{bmatrix}$$

Applied: Total Sales by Matrix Addition

Three stores (A, B, C) each report sales of 3 products over 6 days (M–S). Add the matrices to get total sales per product, per day.

Store A

$$\begin{bmatrix} 40 & 30 & 81 & 0 & 21 & 47 \\ 0 & 12 & 78 & 50 & 50 & 96 \\ 10 & 0 & 0 & 27 & 43 & 78 \end{bmatrix} +$$

Store B

$$\begin{bmatrix} 20 & 19 & 18 & 1 & 12 & 74 \\ 2 & 21 & 87 & 50 & 49 & 96 \\ 10 & 1 & 2 & 72 & 34 & 87 \end{bmatrix} +$$

Store C

$$\begin{bmatrix} 20 & 3 & 18 & 1 & 12 & 45 \\ 1 & 12 & 78 & 5 & 0 & 6 \\ 1 & 2 & 3 & 6 & 8 & 99 \end{bmatrix}$$

Total sales (each entry is a sum across A, B, C):

$$\begin{bmatrix} 80 & 52 & 117 & 2 & 45 & 166 \\ 3 & 45 & 243 & 105 & 99 & 198 \\ 21 & 3 & 5 & 105 & 85 & 264 \end{bmatrix}$$

Equal Matrices

Two matrices are equal if they have the **same size** and their **corresponding entries are equal**.

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$$

$$A \neq B, \qquad A = C$$

A and B differ in one entry, so A ≠ B, even though they're the same size. A and C match entry-for-entry, so A = C.

A System, Rewritten as a Matrix Equation

Recall our general linear system. By the definition of equal matrices, it can be written as a single matrix equation:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

we designate these three matrices A, x, and b, respectively — giving the single equation:

$$Ax = b$$

A is the **coefficient matrix** — columns of A must match rows of x (both = n).

When Can You Multiply Two Matrices?

A and B can be multiplied if the number of **columns of A** equals the number of **rows of B**.

$$A_{m\times n}\ B_{n\times p} = (AB)_{m\times p}$$

"inside" numbers (n, n) must match — that's the compatibility check.

"outside" numbers (m, p) become the size of the product AB.

If the inside numbers don't match, the product simply isn't defined — there's no fallback rule.

Multiplying Matrices — Worked Example

A is 2×3, B is 3×4 — inside numbers match (3 = 3), so AB is defined, and will be 2×4.

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix}$$

Recipe: row i of A · column j of B, entrywise, then add — gives entry c_{ij} of AB.

$$c_{11} = 1(4) + 2(0) + 4(2) = 12$$

Carrying this out for every entry:

$$AB = \begin{bmatrix} 12 & 27 & 30 & 13 \\ 8 & -4 & 26 & 12 \end{bmatrix}$$

Scalar Multiplication

If A is any matrix and c is any scalar, cA is the matrix obtained by **multiplying every entry of A by c**.

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 3 & 1 \end{bmatrix}, \qquad 2A = \begin{bmatrix} 4 & 6 & 8 \\ 2 & 6 & 2 \end{bmatrix}$$

Notice this is entirely different from matrix multiplication — a scalar times a matrix scales every entry; two matrices multiply row-by-column.

Order Matters: AB ≠ BA in General

A and B **commute** if AB = BA — but in general, AB ≠ BA. A quick check:

$$A = \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} -1 & -2 \\ 11 & 4 \end{bmatrix}, \qquad BA = \begin{bmatrix} 3 & 6 \\ -3 & 0 \end{bmatrix}, \qquad AB \neq BA$$

Matrix multiplication is not commutative — a fundamentally different behavior than ordinary number multiplication, and one to watch for all semester.

TRY IT YOURSELF

When Does $(A+B)^2 = A^2 + B^2 + 2AB$?

Question: if A and B are two square matrices of the same size, find the condition such that $(A+B)^2 = A^2 + B^2 + 2AB$. Don't use entries — just matrix symbols.

Solution

$$(A+B)^2 = (A+B)(A+B) = A^2 + AB + BA + B^2$$

$$\text{If } AB = BA: \qquad (A+B)^2 = A^2 + 2AB + B^2$$

The familiar-looking identity from ordinary algebra only survives when A and B commute — a direct consequence of AB ≠ BA in general.

The Associative Law

If A, B, C are matrices such that AB and BC are defined, then **A(BC) = (AB)C**.

Grouping doesn't matter — only the left-to-right order of the matrices does.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$$

$$(AB)C \;=\; A(BC) \;=\; \begin{bmatrix} 18 & 15 \\ 46 & 39 \\ 4 & 3 \end{bmatrix}$$

Both groupings land on the same 3×2 matrix — verify it yourself as practice with the multiplication recipe.

Multiplying by a Diagonal Matrix

Diagonal matrices act like a set of independent scale factors — one per row or per column.

Left multiply

$$D = \operatorname{diag}(d_1, \ldots, d_m) \quad \Rightarrow \quad DA \text{ scales row } i \text{ of } A \text{ by } d_i$$

Right multiply

$$E = \operatorname{diag}(e_1, \ldots, e_n) \quad \Rightarrow \quad AE \text{ scales column } j \text{ of } A \text{ by } e_j$$

Multiplying on the left scales rows; multiplying on the right scales columns.

The Identity Matrix

Set every scale factor to 1: the diagonal matrix **diag(1, 1, …, 1)** is the identity matrix, denoted I_n or just I.

$$I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

For any matrix A of order n:

$$AI_n = I_n A = A$$

I_n commutes with every matrix it multiplies — the matrix analogue of the number 1.

Inverse Matrices

If Q and A are square matrices of the same order n, Q is an **inverse** of A if and only if:

$$QA = AQ = I_n = I \qquad \Longleftrightarrow \qquad Q = A^{-1}$$

Q is written A^{-1}.

Nonsingular / Invertible

A has an inverse.

Singular

No inverse of A exists.

RECAP

Key Takeaways — Matrices

A matrix

a rectangular array of entries; size = rows × columns; square when rows = columns.

Equality & addition

same size required for both; add entry-wise; scalar multiplication scales every entry.

Ax = b

any linear system is a single matrix equation — the coefficient matrix times the unknown vector.

Multiplication

defined only when inside dimensions match; row · column, entry-wise, then sum.

Not commutative

AB ≠ BA in general — order matters, unlike ordinary numbers.

Identity & inverse

I is the matrix analogue of 1; A^{-1} exists only when A is nonsingular.

NEXT TIME

Looking Ahead

Lecture 2

Matrix Algebra & EROs

- Matrix properties and theorems
- Which matrices are invertible?
- Row-reducing [A | I] to find A^{-1}
- Connecting back to solving Ax = b

Homework 1

Refer to OneNote and CANVAS

This does not need to be submitted, but it essential for practice and recap of concepts covered today

Reflection

If AB ≠ BA still feels strange, that's normal — sit with a 2×2 example until the asymmetry feels obvious, not surprising.

Questions?

Systems of Linear Equations & Matrices · Lecture 1 complete

NEXT: MATRIX PROPERTIES & THEOREM, INVERSES